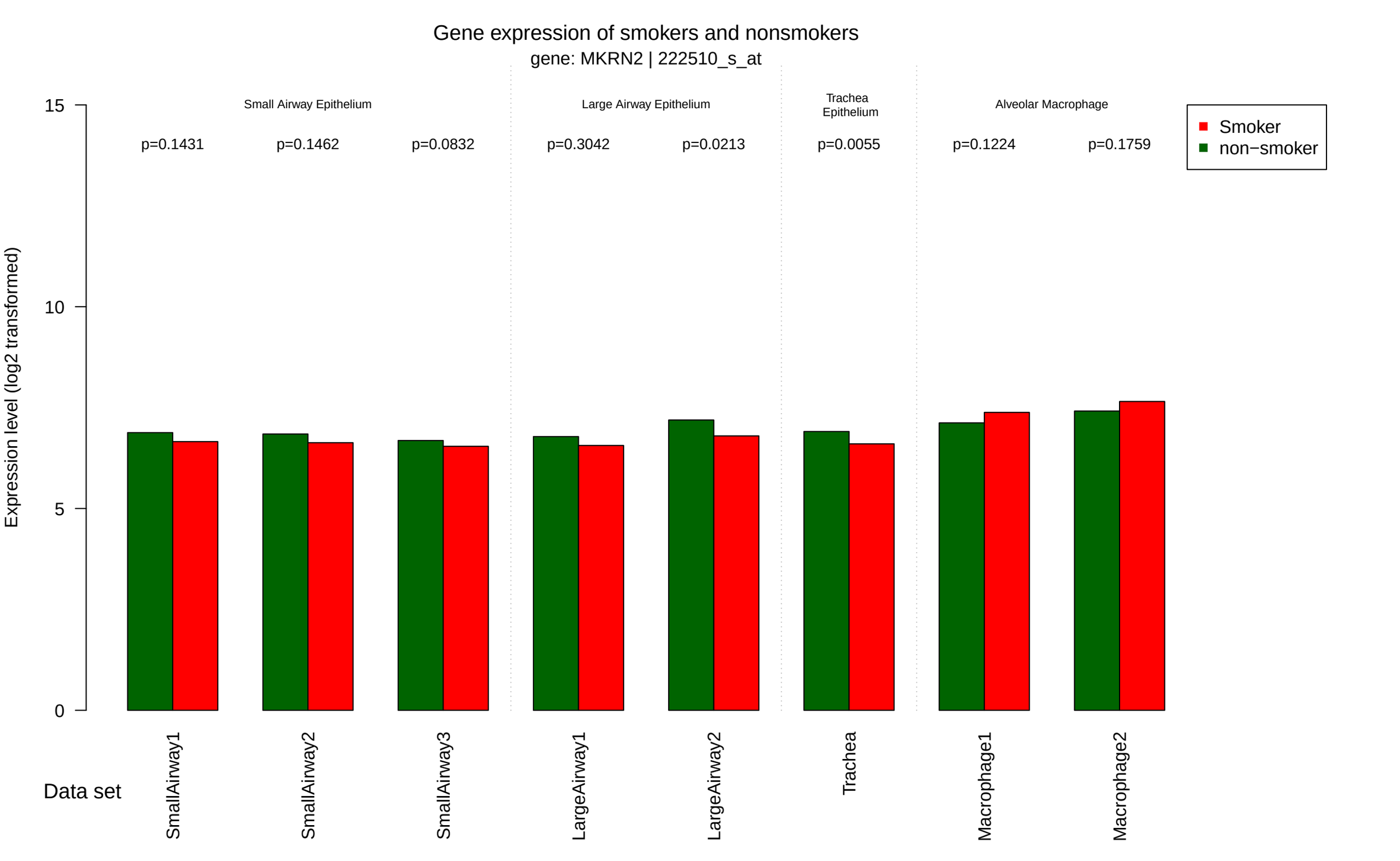
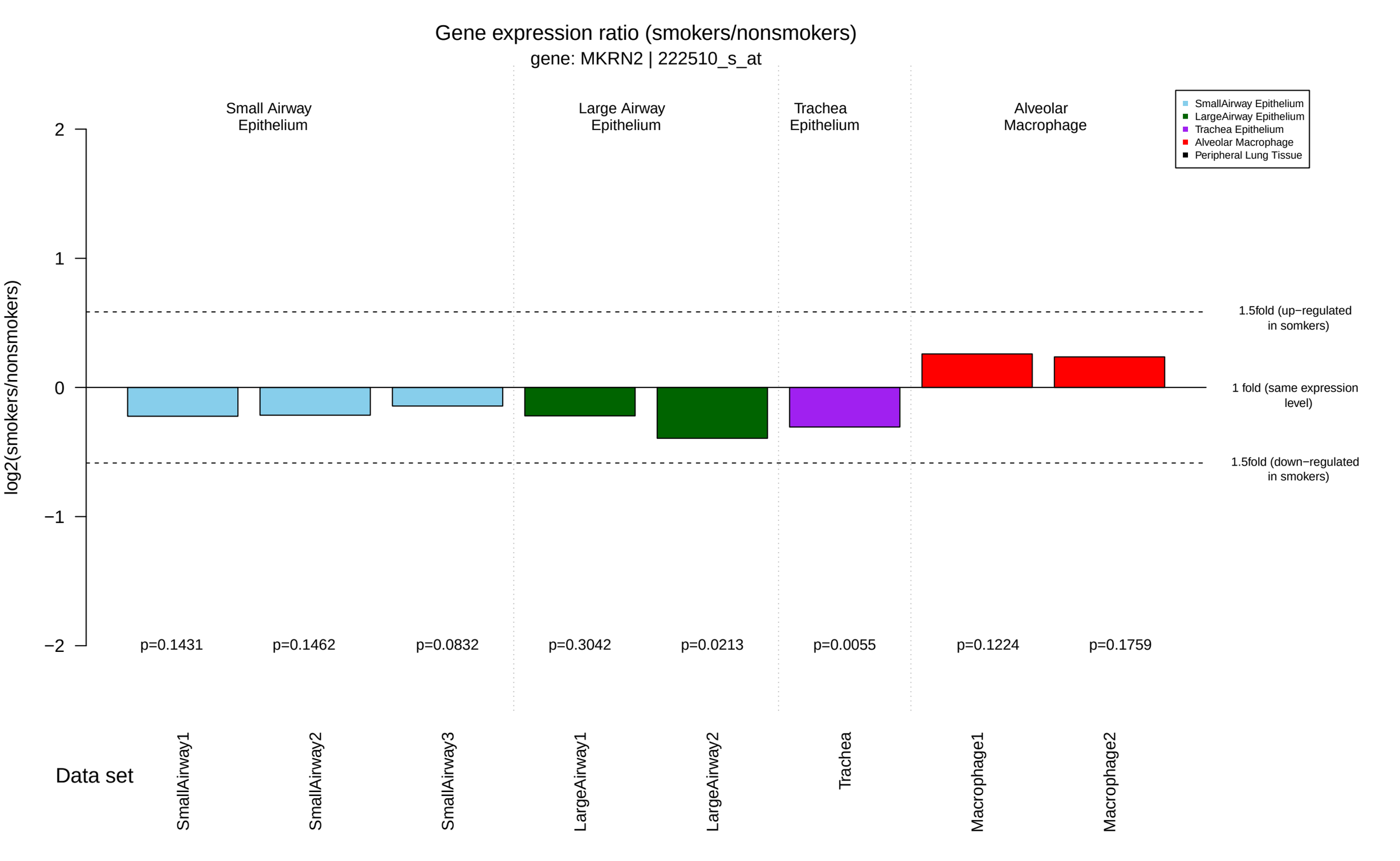
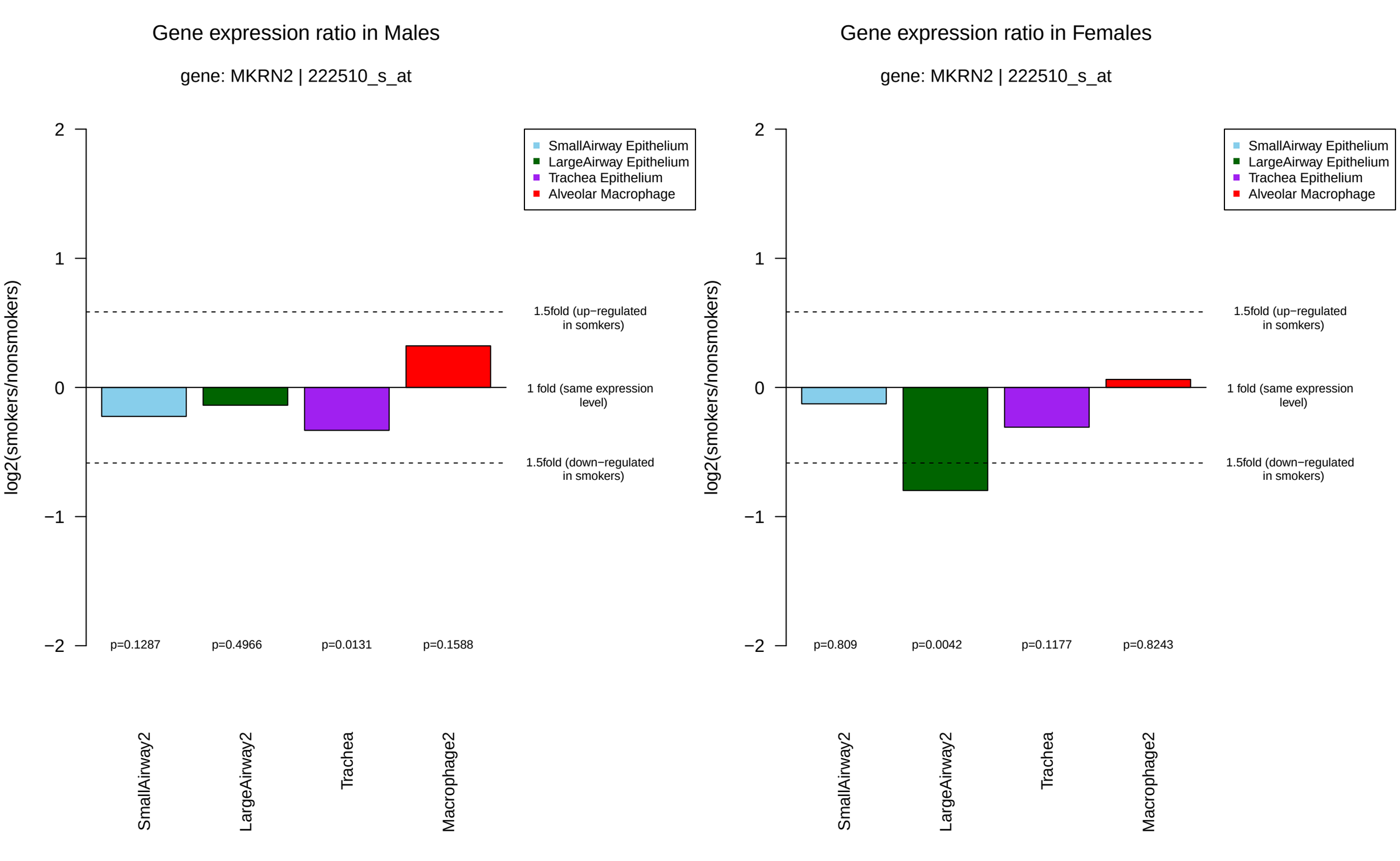
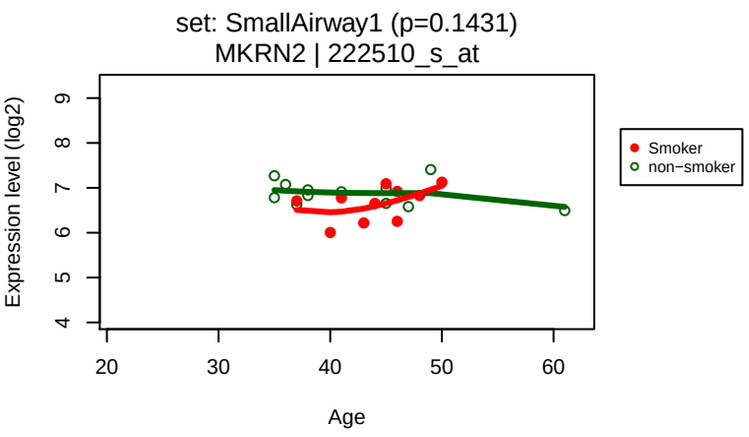
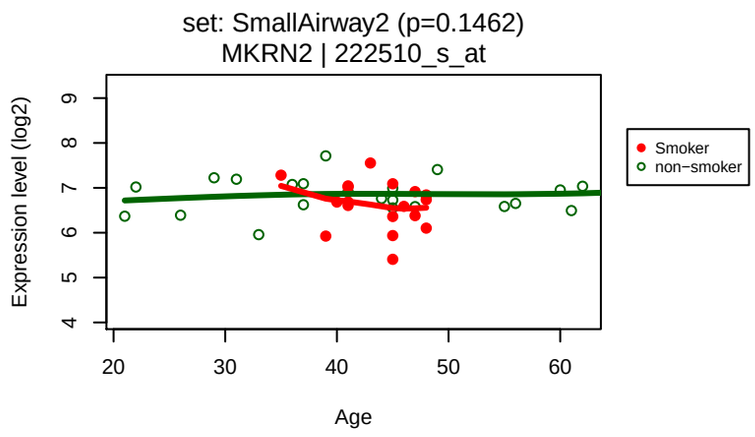
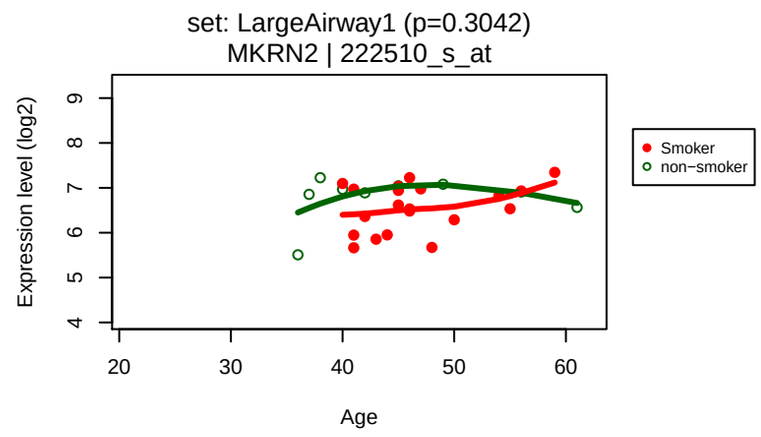

Gene expression of smokers and nonsmokers
gene: MKRN2 | 222510_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1431
p=0.1462
p=0.0832
p=0.3042
p=0.0213
p=0.0055
p=0.1224
p=0.1759
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

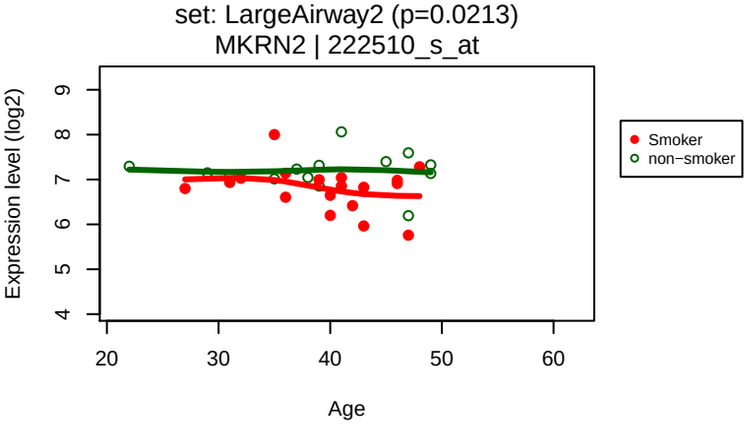

Gene expression ratio (smokers/nonsmokers)
gene: MKRN2 | 222510_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1431
p=0.1462
p=0.0832
p=0.3042
p=0.0213
p=0.0055
p=0.1224
p=0.1759
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

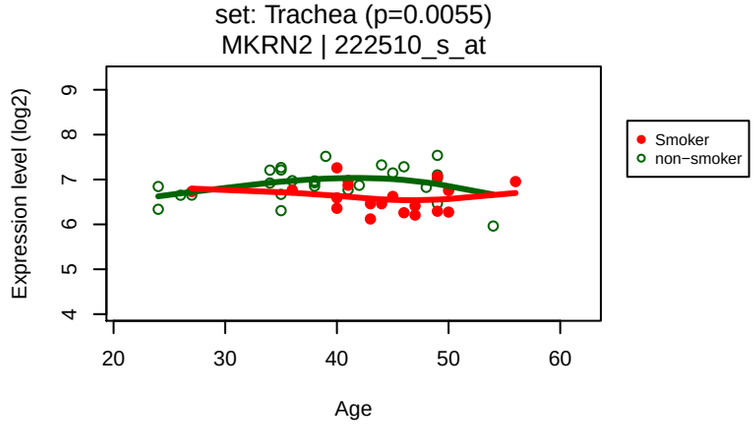

Gene expression ratio in Males
gene: MKRN2 | 222510_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1287
p=0.4966
p=0.0131
p=0.1588
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MKRN2 | 222510_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.809
p=0.0042
p=0.1177
p=0.8243
SmallAirway2
LargeAirway2
Trachea
Macrophage2

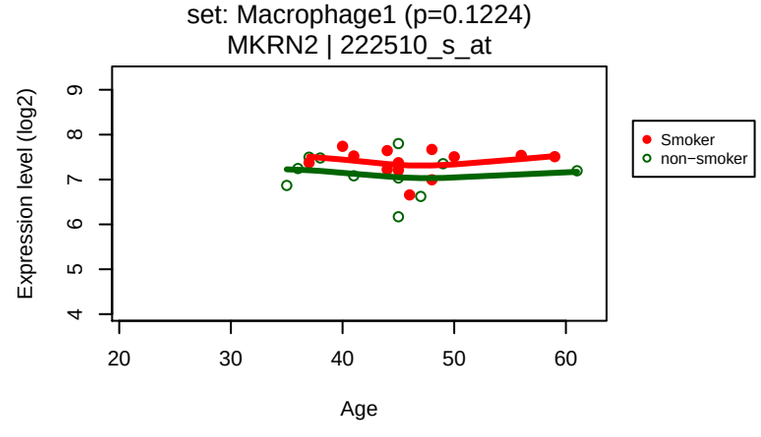

set: SmallAirway1 (p=0.1431)
MKRN2 | 222510_s_at
Expression level (log2)
Age
Smoker
non−smoker

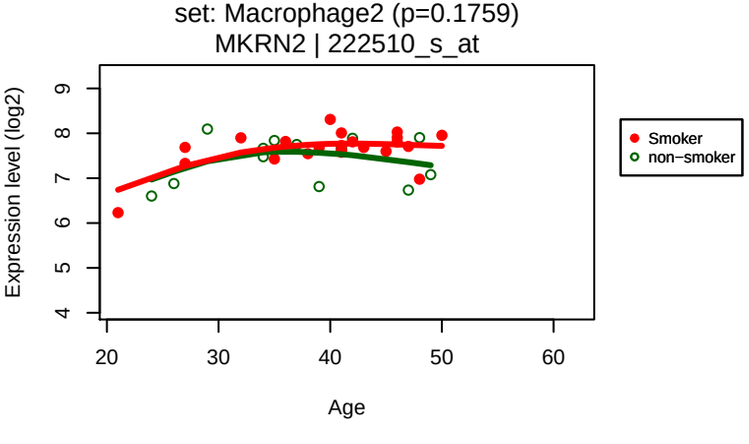

set: SmallAirway2 (p=0.1462)
MKRN2 | 222510_s_at
Expression level (log2)
Age
Smoker
non−smoker

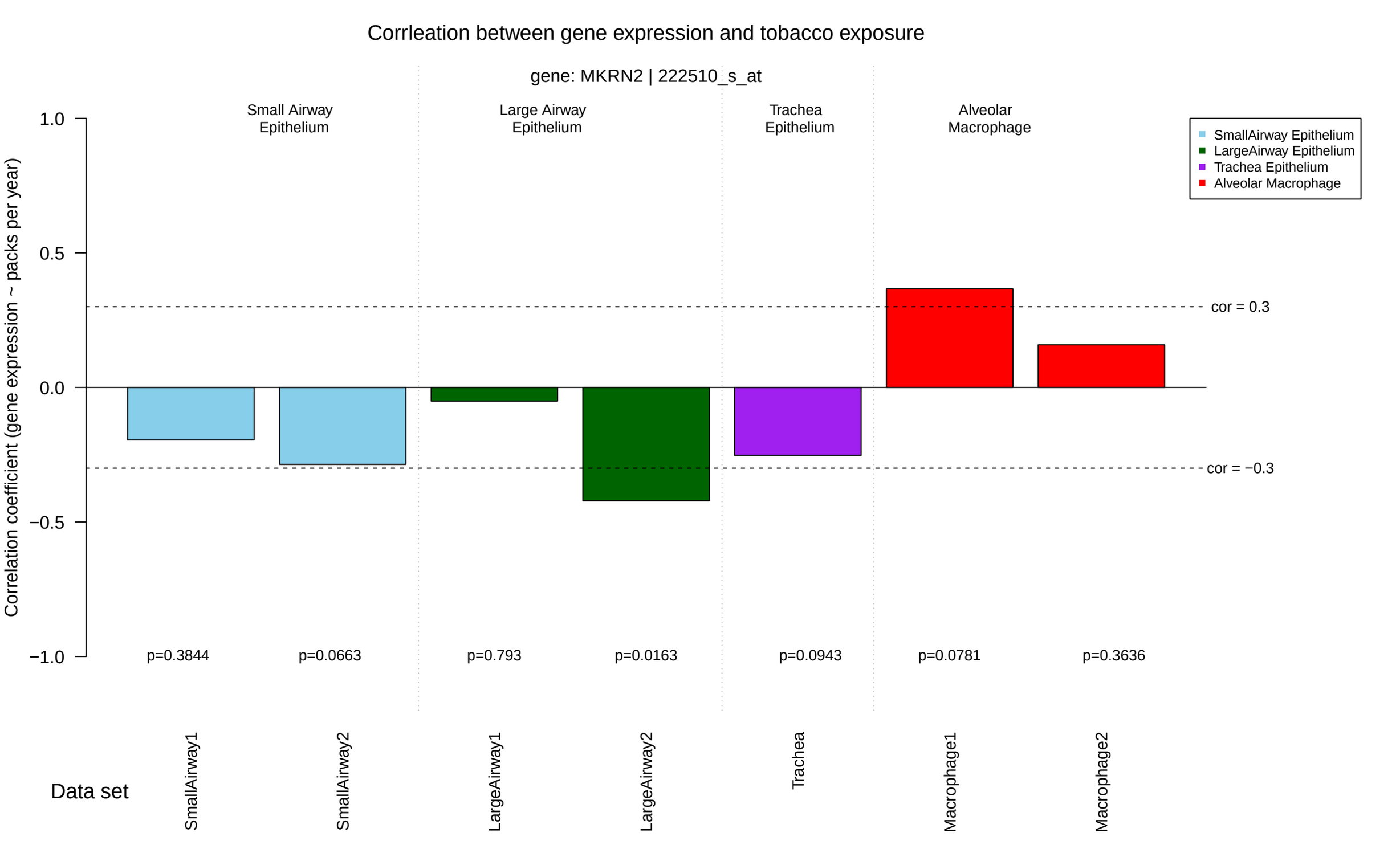

set: LargeAirway1 (p=0.3042)
MKRN2 | 222510_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0213)
MKRN2 | 222510_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0055)
MKRN2 | 222510_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1224)
MKRN2 | 222510_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1759)
MKRN2 | 222510_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MKRN2 | 222510_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3844
p=0.0663
p=0.793
p=0.0163
p=0.0943
p=0.0781
p=0.3636
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2